



Paolo Forin

PhD in Neurorobotics

27 October 1998

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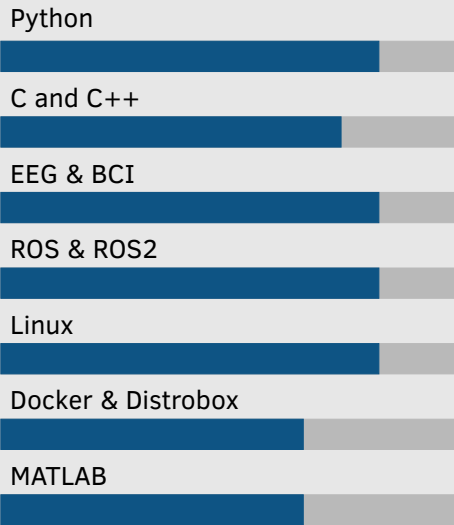
github.com/PaoloFori

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About me

PhD candidate at IASLab (UniPD). I build real-time BCI systems based on Motor Imagery and Covert VisuoSpatial Attention — from collaborative robotic arm control to GMM-based attentional gating. Currently visiting researcher in Lisbon integrating BCI with VR. Strong background in C++, Python, ROS2, and Docker.

Skills



English★5 Italian★6

(*)[The skill scale is from 0 (Fundamental Awareness) to 6 (Expert).]

Education

- 2023–now PhD in Neurorobotics & BMI University of Padova
IASLab – Intelligent Autonomous Systems Laboratory. Research on endogenous BCI paradigms (Motor Imagery & CVSA) for real-time robotic control, collaborative BCI, and VR integration.
- 2021–2023 M.Sc. Computer Engineering – 110/110L University of Padova
Curriculum: Robotics & AI. Thesis: Dynamic vs. exponential control for online 2-class Motor Imagery BCI.
- 2018–2021 B.Sc. Computer Engineering – 102/110 University of Padova
Thesis: Kubernetes-based orchestrator for theatre ticketing system.

Experience

- Feb–Sep 2026 Visiting Researcher Lisbon, Portugal
Integration of BCI systems (MI + CVSA) with immersive Virtual Reality using VRStream.
- 2023–now PhD Researcher – IASLab University of Padova
Real-time BCI development. Contributor to ROSNeuro and ROS2Neuro open-source ecosystems.
- 2020 Internship Lynx S.P.A., Padova
Backend for a theatre reservation app: Java, Spring Boot, Swagger UI, MongoDB, JaCoCo.
- 2017–2022 Private Tutor Padova
Math, physics and science for high school students.

Publications

- P. Forin et al., “Attentional Gating for CVSA-Based BCI Using GMM,” *Graz BCI Conf. 2026 – accepted*.
- P. Forin, S. Tortora, E. Menegatti, L. Tonin, “Towards a shift and sustain-based decoding architecture for CVSA BMI in a natural environment,” *IEEE, 2025*.
- P. Forin, G. Beraldo, S. Tortora, L. Tonin, “Optimization and evaluation of the control framework for BMI,” *IEEE CASE 2024, Bari*.
- A. Palatella, P. Forin, S. Tortora, E. Menegatti, L. Tonin, “An alternative training protocol for a MI BMI based on a collaborative approach,” *Graz BCI Conf. 2024. – 3rd place ICRA 2025 (poster & EXPO)*.
- G. Beraldo, P. Forin, L. Tonin, “Optimising the continuous control of brain-actuated robotic devices,” *CEUR – AIRO 2023, Udine*.

Projects

- 2024–26 PhD BCI Projects IASLab & Lisbon
CVSA BCI (QDA + mean power, audio feedback) · *cvsa_ws/simple_cvsa*
Collaborative BCI (2 users, 1 robotic arm, shared control)
Attentional Gating (GMM focus detection, CVSA+MI) · *cvsa_ws/ic_cvsa*
BCI in VR (MI + CVSA + VRStream, Lisbon) · *bci_vr_ws*
ML vs DL for MI BCI – real-time comparison (*in progress*)
ROSNeuro & ROS2Neuro contributor · *github.com/rosneuro*
- 2020–23 Degree Projects University of Padova
MI BCI Control (M.Sc. thesis) · *Continuous_control_wheelchair_bmi*
Boat Detector HOG + OpenCV/C++ · *boat_detector*
Vehicle Counting HOG + YOLO/Python · *vehicle_counting_HOG_YOLO*
Theatre Orchestrator Kubernetes + Kafka (B.Sc. thesis) · *Simulation_theater_reservations*
- 2022–24 Extra Projects Personal
Telegram Rental Bot Python + Docker + NAS · *telegram-rental-bot*
Pykemon Pokémon battle simulator in Python (PhD course) · *pykemon*